
Transformer Genre Classification with Interpretability Using Attention Vectors

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Abstract

Through this project, we seek to apply deep learning models to music, to glean insights about genre-defining characteristics. We began by building a genre classifier using a transformer model. We began with a simple model with 2 layers and 4 attention heads, then quickly improved to a 4 layer and 4 head per layer model. We tokenized our audio files using mel spectrograms, a time-frequency representation of audio that show how the frequency content evolves over time; through these we managed to compress each 30 second audio into a (1290,128) tensor, which we then used to train and test our model. We managed to reach a xx accuracy, xx precision, xx recall on our unseen data. Finally, we extracted the attention vectors from our model to analyze which characteristics are particularly important in predicting what genre an audio is.

1 Introduction and Motivation

1.1 What problem are you solving?

This project applies deep learning to music to uncover genre-defining characteristics and improve genre classification accuracy. Our first objective is to build a classifier that can predict the genre of a song based on its audio features, using transformer-based models well-suited for sequential data like music. Beyond classification accuracy, we aim to understand why the model makes its predictions by analyzing the attention vectors. This allows us to identify which parts of a song are most influential in determining its genre, offering deeper insight into the acoustic patterns that distinguish genres. Ultimately, our goal is not only to develop a high-performing classifier but also to advance interpretability by revealing the specific audio segments that contribute most to genre identification.

1.2 Why is this problem important or interesting?

This project is compelling because it explores music as structured, sequential data. While we often work with structured tabular data, music shares key properties with language: both are sequences where meaning emerges from temporal patterns, meaning that many of the concepts found in natural language processing (NLP) may also apply here. Just like how NLP uses attention and transformers to understand the underlying structure of text, we will use them to investigate whether genre can be predicted based on patterns captured from raw audio features. The most novel aspect of our work is not just genre classification, but using the model's attention vectors to analyze which parts of a song contribute most to genre identification.

The most novel aspect of our work is not just genre classification, but leveraging the model's attention vectors to analyze which parts of a song contribute most to genre identification. In NLP, attention reveals which words are most important to meaning; we extend this idea to music, asking whether

certain song segments play a particularly critical role in defining genre. Our project demonstrates one way to apply well-known machine learning principles to a less conventional data type, and shows how transformer-based models can shed light on the inner workings of raw audio data—an area with significant commercial and academic relevance.

1.3 What real-world or research need does it address?

Genre classification has widespread applications in media, entertainment, and information retrieval. The most prominent use case for our model lies in the music industry. Genre classifiers allow streaming services to group music and personalize recommendations, generate mood- or activity-based playlists, and maintain accurate metadata across millions of tracks without requiring manual labeling.

Beyond music, genre classification plays a vital role in video and news publishing platforms, where it supports content categorization and recommendation algorithms that improve the experience for users. In academic and research contexts, genre classification helps analyze patterns across time periods, artists, and media types, and contributes to our understanding of how people interact with different forms of content.

In the context of music, two of the largest streaming services—Apple Music and Spotify—use genre classification as a important tool in building user-tailored content discovery systems. Given this, there is strong demand for high-performing genre classification models. Moreover, if we can interpret which parts of a song are most influential in determining its genre, this could offer valuable insights into how genre conventions evolve and which auditory cues fans of particular styles enjoy the most.

2 Background and Related Work

Genre classification has been attempted in the past, most commonly on the GTZAN dataset. When a CNN model was trained end-to-end to predict the genre of a music signal, an accuracy of 76% was achieved [3]. Another 1D CNN architecture achieved an accuracy of 81% [1]. This paper extracted data using digital signal processing techniques and then used their CNN to perform the genre classification task. Ensemble learning techniques have also been used. For example, LSTM and CNN conjunctions have been used in the GTZAN data set to perform genre classification, with one model achieving a classification accuracy of 82% [2]. Another hybrid model using ResNet and the Gated Recurrent Unit achieved a genre classification accuracy of 90% [5]. Even more, graph neural networks, using information at an artist level in order to link tracks, have been used for this task and achieved a final F1 score of 79% [4].

While the previous literature has been focused on achieving higher accuracy scores, there have been few serious attempts at explaining which parts of the songs are the most important to classification or most genre defining. In [2], they use subcomponent level analysis in order to improve the accuracy of their genre classifier but the authors do not dive deeply into what parts of the audio file are most important in classifying a song in a certain genre. In [2], the author attempts to use graph structure based on artist similarity in order to define genres, which inherently has some degree of interpretability. Still, they do not attempt to open the black box of their model, leaving much to be understood in seeing which features are the most important for genre classification.

In this work, we aim to fill this hole in interpretability. Through the use of attention vector analysis, we aim to find the most important time in the audio file in terms of classifying the song. We chose to analyze the attention vectors, as these will reveal which parts of the song are the most impactful, leading us to see which timestamps will have the largest effect on the final classification. At the same time, we hope to build a model that performs with higher accuracy than previous ones through the tokenization of the audio data. By doing this, we aim to improve on the accuracy of the aforementioned models, while also offering a deeper gaze into the mechanisms by which one would classify genres.

3 Methods / Model

The most challenging part of our project was the tokenization process. Each audio file was a 30-second WAV file, containing roughly 665,000 amplitude values. This raw data was far too large for

our model to process directly, so we needed to compress it while preserving its sequential nature. We explored two approaches: the first was mel spectrograms. Spectrograms convert audio into the frequency domain, showing how frequencies vary over time—a format well-suited to music recognition tasks. A mel spectrogram maps frequencies onto the mel scale, which better reflects human pitch perception. Using the librosa library, we transformed each audio file (665,000 samples) into a (1290, 128) tensor, where each row represents a 23-millisecond slice of audio, maintaining the sequence structure in a format that the transformer can handle effectively.

The second method used wav2vec, a pretrained audio transformer developed by Facebook. We used this model to generate (128, 1) embeddings for each audio file and fed these into our classifier. While these embeddings performed well, they lost the sequential aspect of the data, which made it impossible to interpret attention vectors in a meaningful way.

Another challenge was the limited size of our dataset. We used the GTZAN Genre Collection, a well-known benchmark dataset for music classification, containing 1000 30-second audio clips across 10 genres. Although clean and widely used, the dataset is relatively small. To address this, we applied data augmentation: each sample had a 50% chance of being augmented with one of four transformations—time stretching, pitch shifting, Gaussian noise addition, or volume scaling.

Our final model is a transformer-based architecture with 4 encoder layers and 4 attention heads per layer, using a model dimension of 256 and a feedforward network size of 512. This design balances model complexity with the small dataset size (5000 samples after augmentation), allowing it to learn meaningful temporal and spectral patterns without overfitting. Each 30-second clip, represented as a (1290, 128) mel spectrogram, is processed with a CLS token and sinusoidal positional encodings; the final CLS embedding is then classified into one of 10 genres. We initially used a 2-layer, 4-head model, which performed well but showed little interesting structure in its attention vectors. This led us to deepen the model slightly to better capture salient audio features.

4 Empirical and/or Theoretical Results

4.1 Key metrics

Accuracy, classification report and confusion matrix

Genre	Precision	Recall	F1-score	Support
Blues	0.92	0.97	0.94	120
Classical	1.00	1.00	1.00	120
Country	0.94	0.93	0.94	120
Disco	0.98	0.93	0.95	120
HipHop	0.89	0.99	0.94	120
Jazz	0.98	0.98	0.98	119
Metal	0.97	0.95	0.96	120
Pop	0.97	0.95	0.96	120
Reggae	0.95	0.95	0.95	120
Rock	0.91	0.89	0.90	120
Accuracy			0.95	1199
Macro avg	0.95	0.95	0.95	1199
Weighted avg	0.95	0.95	0.95	1199

Table 1: Classification report for the genre classifier, showing precision, recall, and F1-score per genre along with overall accuracy.

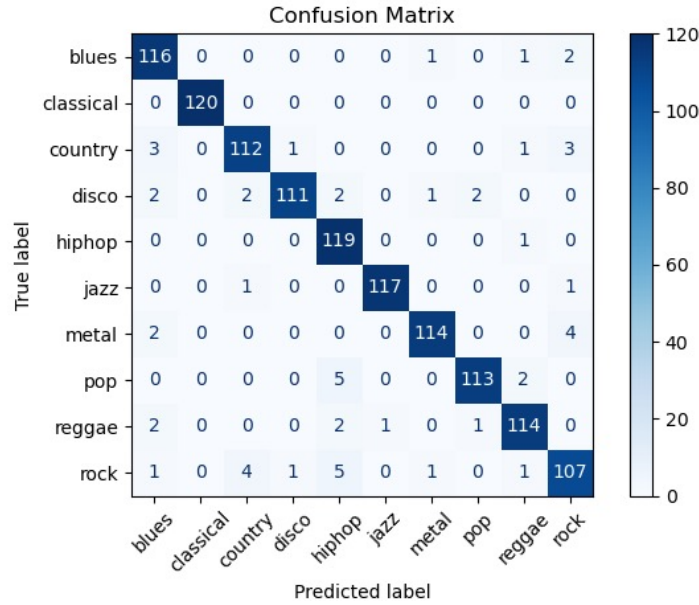


Figure 1: Confusion matrix

It is interesting to observe from the classification report that the model predicts certain genres with near-perfect accuracy, while struggling more with others. For instance, classical music is perfectly classified: all 120 samples are correctly identified, and no other genre is misclassified as classical. This likely reflects that classical music has distinctive structural characteristics that set it apart from other genres. In contrast, the model has more difficulty with pop and rock, which are frequently confused with hip hop. This is intuitive, as these genres share overlapping acoustic features and stylistic elements. Overall, the confusion matrix suggests that the model captures meaningful structural patterns within the music, particularly for genres with more distinct signatures.

Note that the generalization error for the model was quite low. The accuracy on the training set was 99.81% and 95.22% on the test set. While I do believe the model was a overfitting a little bit on the training data, it was still generalizing quite well.

4.2 Attention maps

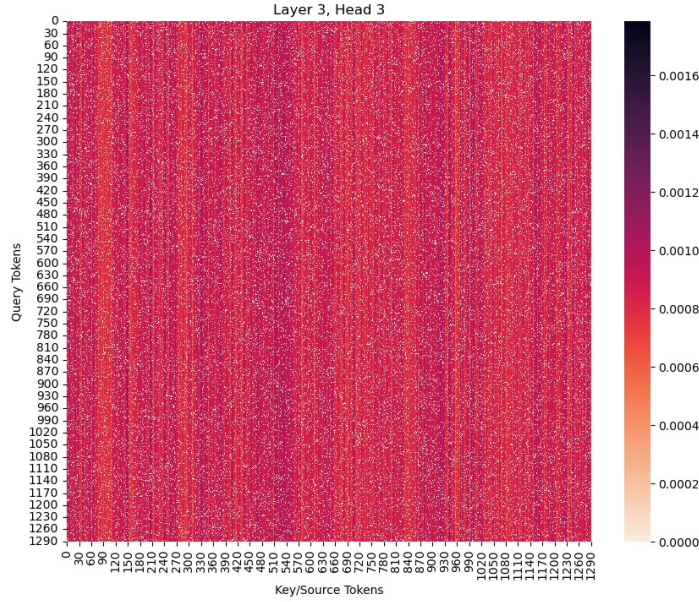


Figure 2: Attention map extracted from layer 3, head 3

Before conducting a per-genre analysis of the attention maps, we first inspected them for a random sample from the test set. Interestingly, across all heads and layers, we consistently observed vertical lines in the attention maps. This pattern indicates that many query tokens are focusing on the same key token. For instance, in the map above, there are elevated attention values around key token 540, corresponding to approximately second 12 in the audio file. This suggests that nearly all time steps in the sequence are assigning significant attention to that moment. Such a pattern implies that this particular part of the song is acting as a global anchor for genre identification. For example, an intro beat or a distinctive rhythmic element at that time point could serve as a strong genre cue, attracting widespread attention from the model.

4.3 CLS attention

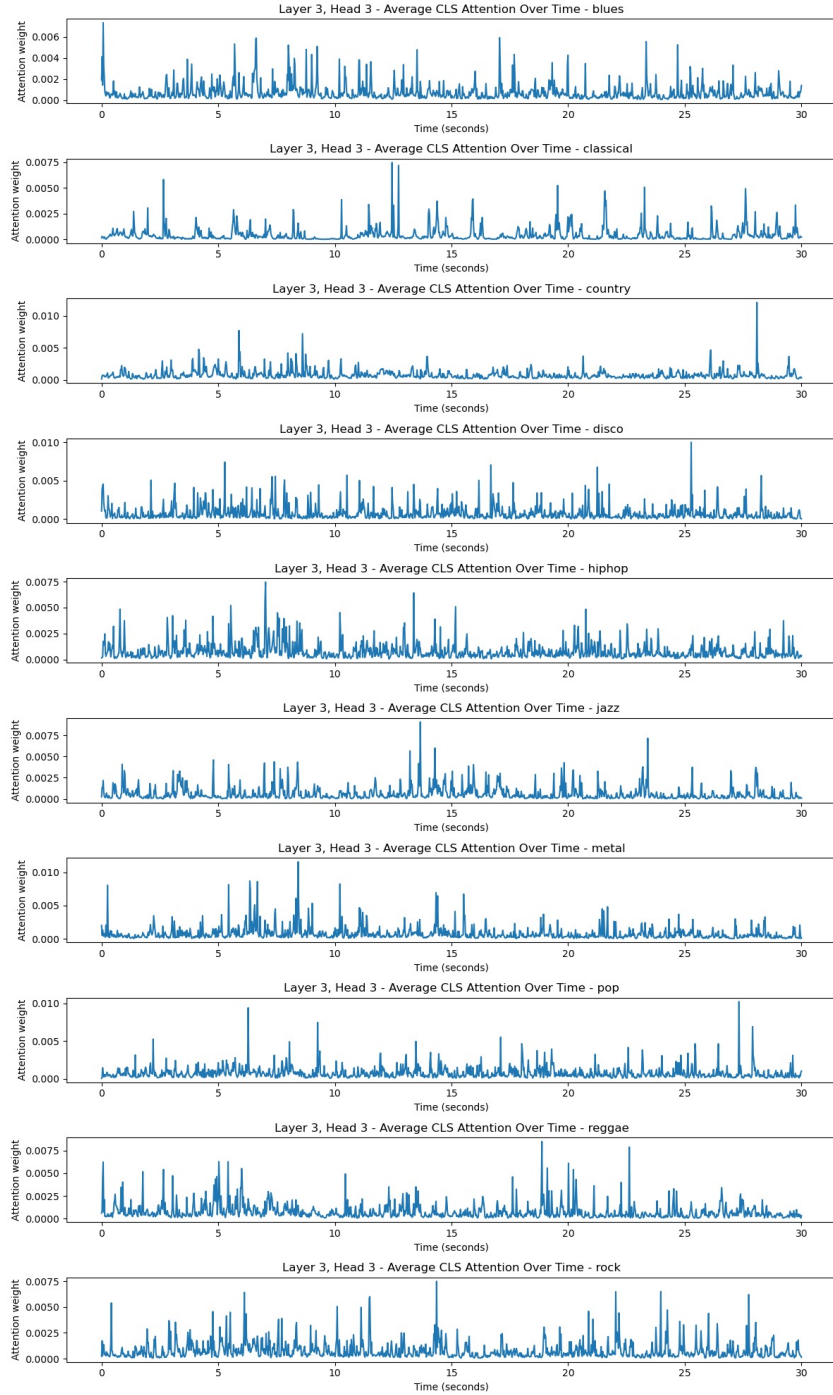


Figure 3: Information extracted from layer 3, attention head 3 per genre

These graphs show the average CLS attention weights over time for each genre, computed from Layer 3, Head 3 of the transformer model. This head and layer were chosen somewhat arbitrarily, but we focused on a deeper layer because it is typically at that point in the model where higher-level features have been extracted. Each plot represents the mean attention that the CLS token assigns to each time

step across all test samples of a given genre. The x-axis corresponds to time in seconds, while the y-axis reflects the average attention weight.

In transformer models, the CLS (classification) token is a special learnable embedding prepended to the input sequence. During training, this token is designed to aggregate information from the entire sequence through the self-attention mechanism, effectively acting as a summary representation of the input. At the output, the final embedding of the CLS token is passed to the classifier to make the genre prediction. Because the CLS token is trained to concentrate the most relevant features for classification, analyzing the attention it assigns to different time steps gives us insight into which parts of the input the model deems most important for its genre classification decision.

This visualization lets us investigate whether certain temporal regions of songs consistently hold more weight in genre classification. Peaks in the attention weights highlight moments in the audio that the model considers especially informative. For example, in the 'blues' genre, we observe a strong peak at the very beginning of the song, suggesting that the opening note is a key genre signal. Conversely, for 'country,' there's a distinctive peak near the end of the clip (around 28 seconds), which appears consistently across samples.

We can extend this analysis to other heads and layers to gain a richer understanding of what the model focuses on. A next step could involve revisiting the audio clips themselves, aligning the high-attention timestamps with specific notes or rhythms to identify the actual musical features driving the model's predictions.

Comparing these graphs across genres also reveals interesting patterns: some genres (like classical or jazz) show more distributed attention, implying that genre cues are spread throughout the song, while others (like disco or hip hop) show sharp, localized spikes, indicating key moments that dominate the genre decision. This supports the idea that the model learns genre-specific temporal signatures and leverages them as anchors for classification.

5 Conclusions and Future Work

In this project, we successfully developed a transformer-based classifier capable of predicting the genre of a song with high accuracy, precision, and recall, outperforming some benchmarks we found in existing research using the same GTZAN dataset. This demonstrates that even with a relatively small dataset, a carefully tuned transformer architecture can capture meaningful temporal and spectral patterns in audio data.

Beyond classification performance, the most novel aspect of our work was analyzing the attention vectors to understand which parts of a song are most influential in determining its genre. Our attention visualizations revealed that specific moments in the audio clips carry disproportionate importance for genre prediction. This suggests that the model isn't just "memorizing" the data but is honing in on distinctive temporal features, which may correspond to genre-defining musical cues like intros, instrument solos, or rhythmic drops. One thing we also did notice was that the attention values themselves remained quite low throughout. It's unclear whether this is a consequence of the inherent scale of the wav audio inputs or a reflection of the model's difficulty in identifying strong, dominant patterns within the audio data. Regardless, it is an interesting phenomenon that warrants further investigation to better understand how attention behaves in the context of raw audio.

For future work, a natural next step would be to dive deeper into these high-attention regions. By aligning the timestamps of attention peaks with the actual audio, we could attempt to identify the precise musical elements (pitch changes, beat drops, rhythm shifts) that the model is focusing on, adding a new layer of interpretability to genre classification.

Another exciting potential direction would be to leverage the insights gained from attention analysis to inform generative modeling. For instance, once we understand which patterns the model associates with each genre, we could explore using generative models to synthesize new music that embodies those genre-specific characteristics. We unfortunately did not have the time to implement that this time, but are excited about the potential idea!

References

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